

Scaling a Vision Transformer Block on SparkRV: an Open-Source Many-Core Digital Neuromorphic Processor

Mohamed Saber

Master in Embedded Artificial Intelligence, Ibn Zohr University

Supervised by Dr. Amirreza Yousefzadeh (University of Twente)
and Prof. Yasmine Lamari (Ibn Zohr University)

Abstract

We present the first mapping of a standard (non-spiking) DeiT-Tiny transformer block onto a RISC-V based digital neuromorphic processor, and characterise the Performance, Power, and Area (PPA) trade-offs of scaling it across a multi-core neuromorphic system. Using a hardware-validated baseline, column-row matrix partitioning, and head-splitting, we show that asymmetric multi-core scaling achieves a $14.73\times$ throughput improvement over the single-core baseline at near-constant energy per token, while Multi-Head Self-Attention (MHSA) remains the dominant source of Network-on-Chip (NoC) traffic.

1 Motivation and Gap

Deploying Vision Transformers (ViTs) on resource-constrained edge devices demands hardware that reconciles dense attention workloads with tight energy budgets. Neuromorphic processors offer event-driven efficiency but impose strict per-core memory and compute limits. Prior work has largely sidestepped standard transformers by converting artificial neural networks to spiking equivalents or recasting attention into binary spiking operations. This leaves the direct mapping of a conventional transformer block onto neuromorphic hardware unexamined, and raises the question of whether the standard attention mechanism is compatible with the constraints of distributed, event-driven execution across multiple cores.

2 Methodology

We map a single DeiT-Tiny transformer block ($N=198$ tokens, $D=192$, 3 heads) onto the SparkRV processor operating at 500 MHz. We first establish a hardware-validated baseline: unsupported transcendental operations (softmax, GELU) are approximated on-core, and every sub-operation is verified against a C golden model to a $1e-6$ floating-point tolerance, guaranteeing functional equivalence with the reference model. We reduce the single-core activation footprint by 29.0% (1,449 KiB) through in-place softmax and GELU buffer reuse. For multi-core deployment we distribute the block via spatial mapping using column-row matrix partitioning and head-splitting, and compare two strategies: a *balanced* 31-core configuration (the memory floor below which the block does not fit given 256 KiB per core) and an *asymmetric* 43-core configuration that allocates additional cores to the compute-dominant FC1 projection.

3 Results

All PPA figures are reported for a GlobalFoundries 22nm process (typical corner, 0.5 V, 25°C) using reduced threshold voltage without back-biasing.

Setup	Throughput (tok/s)	Energy (mJ/token)	Area (mm ²)	Speedup
1-core	0.026	2.497	6.3	1.00×
31-core (balanced)	0.255	2.496	11.6	9.81×
43-core (asymmetric)	0.383	2.496	16.2	14.73×

Table 1: PPA trade-offs across core-count configurations.

On a single core, the block requires 3,772 GCycles and 5 MB of data memory, yielding a throughput of 0.026 tokens/s at 2.497 mJ/token, the unaccelerated MLP GEMMs dominate, with FC1 alone accounting for 82.1% of block latency. Under multi-core scaling, MHSA contributes less than 1% of compute latency yet generates 63.79% of all NoC traffic and roughly three times the communication energy of the MLP, identifying attention as the structural communication bottleneck. Table 1 summarises the PPA sweep: asymmetric scaling to 43 cores reaches a 14.73× throughput improvement while energy per token remains essentially constant, confirming that parallelisation redistributes fixed compute work to gain throughput rather than energy efficiency.

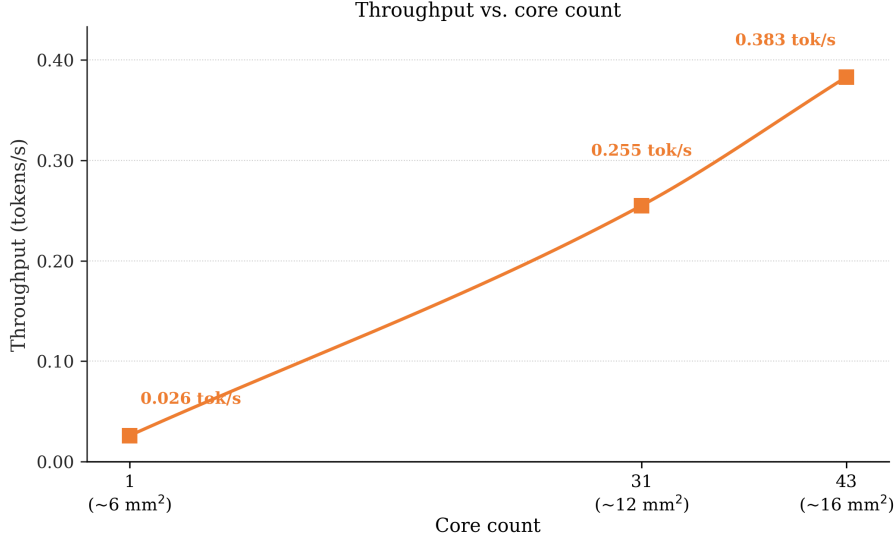


Figure 1: Throughput vs. core count: throughput scales significantly with added cores.

4 Conclusion

These findings establish that a standard vision transformer block can be mapped and executed correctly on distributed neuromorphic hardware, and that throughput scales effectively when parallelisation targets the identified compute bottleneck rather than scaling all layers uniformly. The persistent NoC pressure from all-to-all attention traffic points to hardware multicast support as the principal lever for future improvement. We present this as an initial step toward deploying larger transformer models on event-driven architectures using massive core grids.